Permeation of hydrogen across graphdiyne

Mateo Rodríguez Suárez

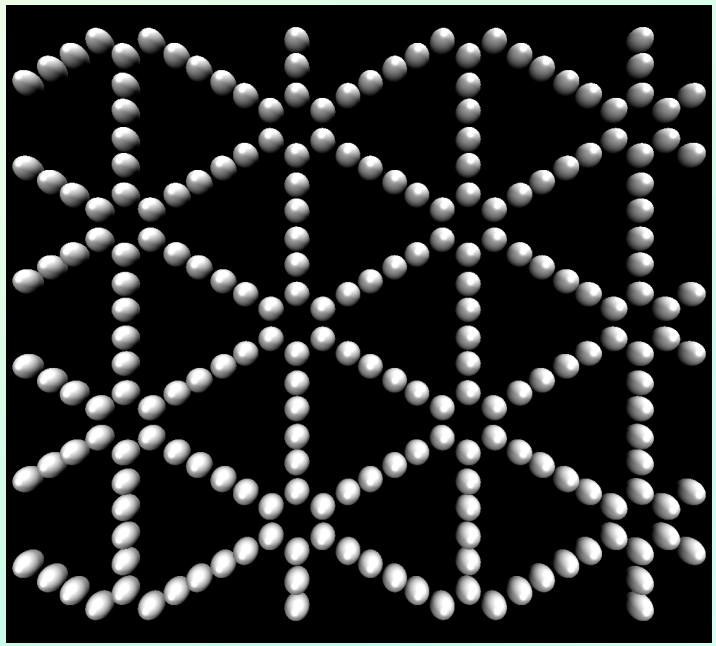
GEFAM 2026 SUMMER SCHOOL ON ATOMIC AND MOLECULAR PHYSICS

31/07/2026

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CONSEJO SUPERIOR DE INVESTIGACIONES CIENTÍFICAS

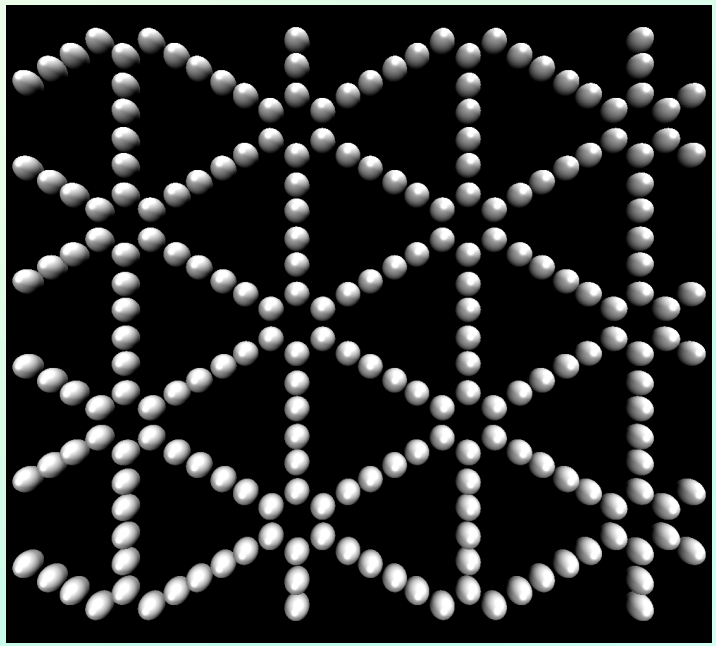
Graphdiyne (GDY)

- Carbon-based nanoporous 2D material
  - One-atom thick
  - Pores of ~ 6.3 Å^2



Graphdiyne (GDY)

- Carbon-based nanoporous 2D material
  - One-atom thick
  - Pores of ~ 6.3 Å^2
- Useful for:
  - Energy storage
  - Catalysis
  - **Gas separation**

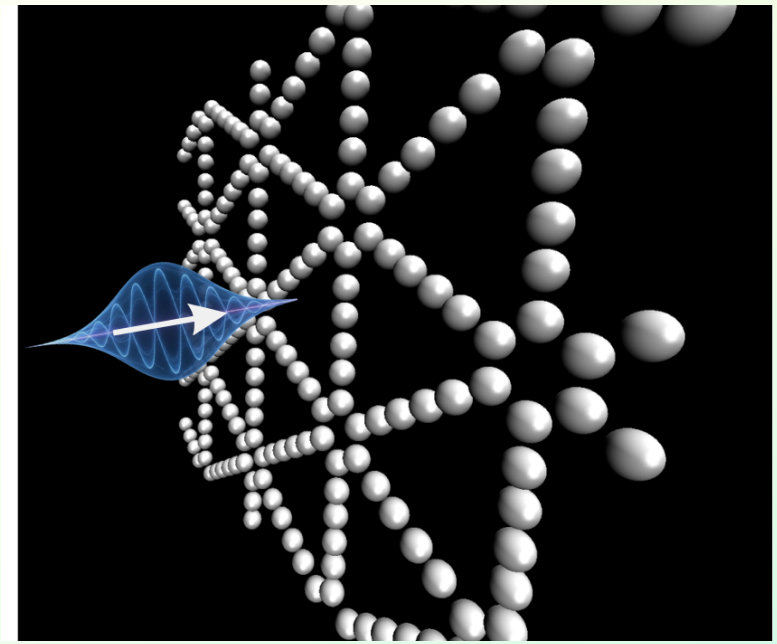


Computational methods

3D-TDWP[1]

Quantum

- Fixed lattice



[1] Lecture 7, José Campos-Martínez

Computational methods

3D-TDWP[1]

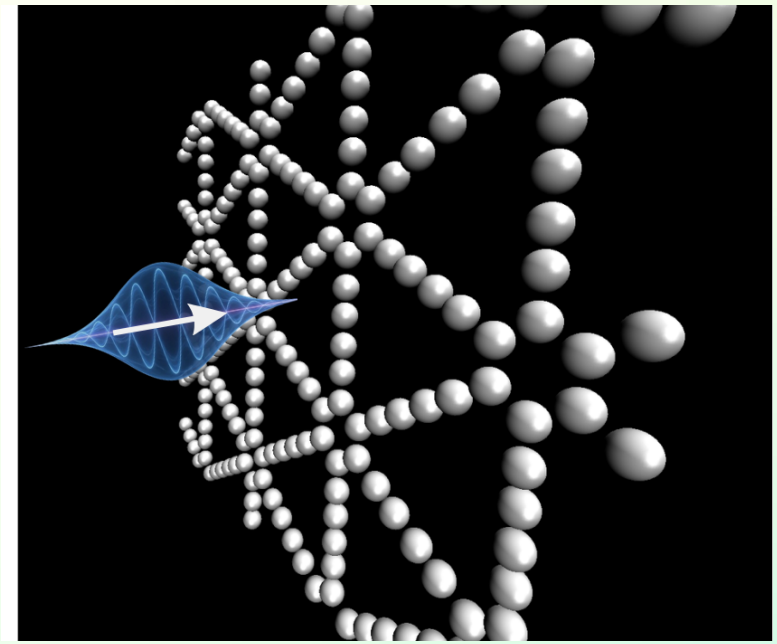
Quantum

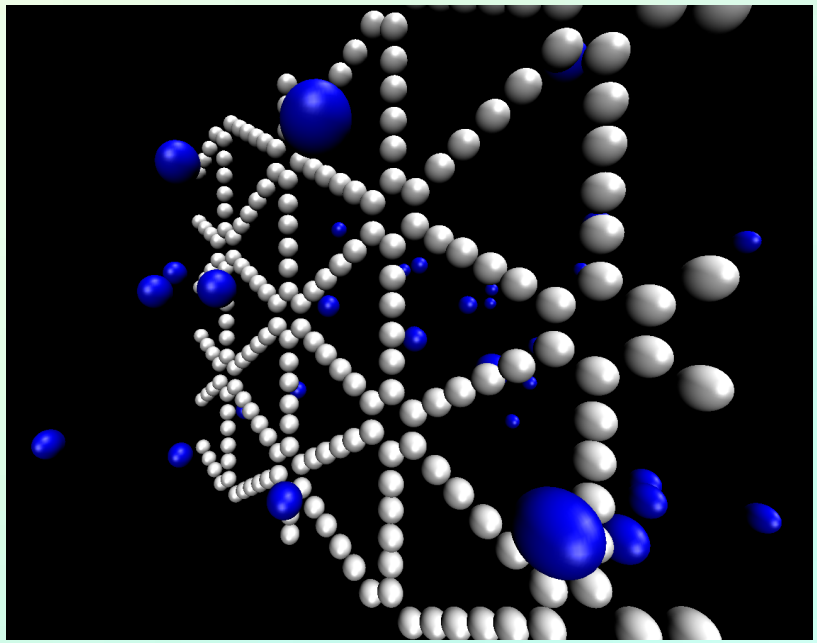
- Fixed lattice

Molecular Dynamics (MD)

Classical

- Fixed or deformable surface





- Permeance: H_2 transport across the GDY layer

[1] Lecture 7, José Campos-Martínez

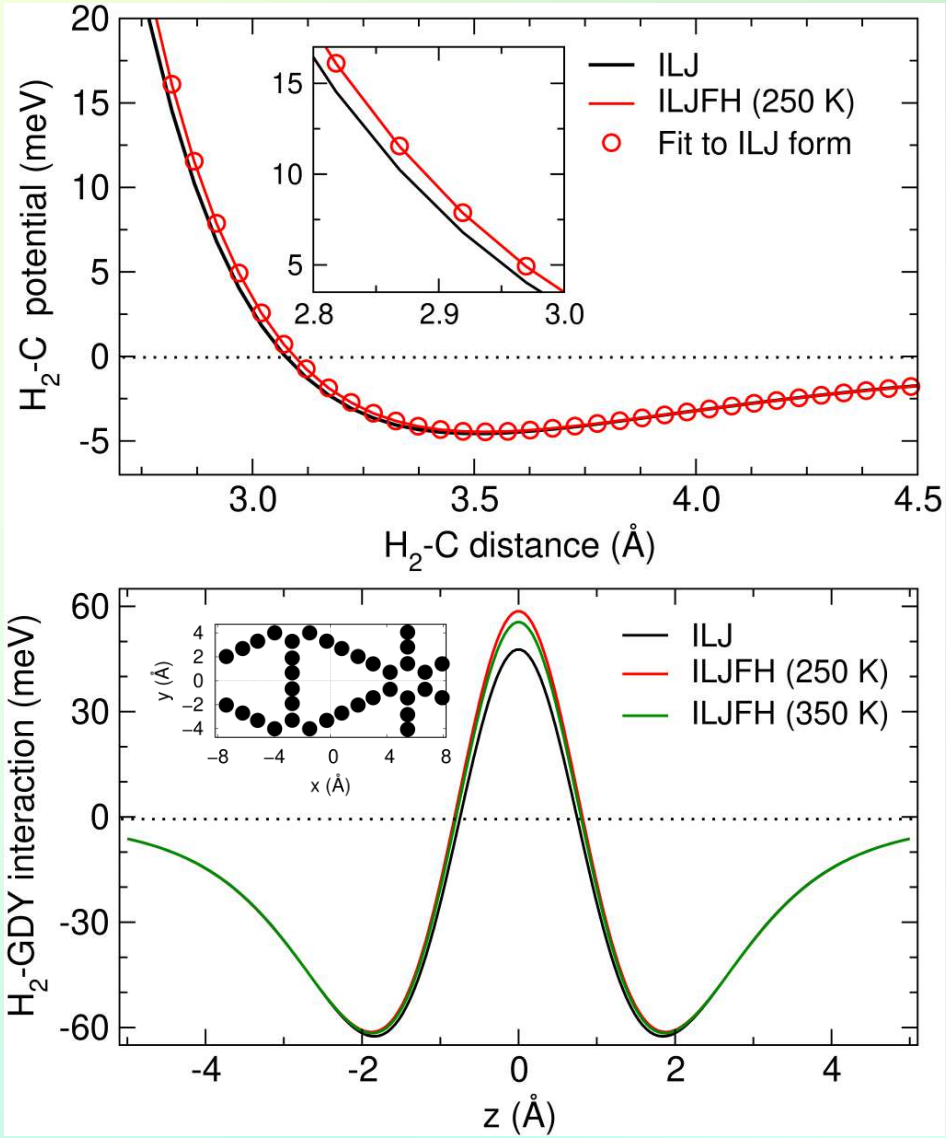
Force fields

- Quantum simulations:
  - Improved Lennard-Jones (ILJ)
- Classical simulations:
  - ILJ
  - ILJ Feynman-Hibbs (ILJFH)

↓

★ Quantum correction
★ Temperature-dependent

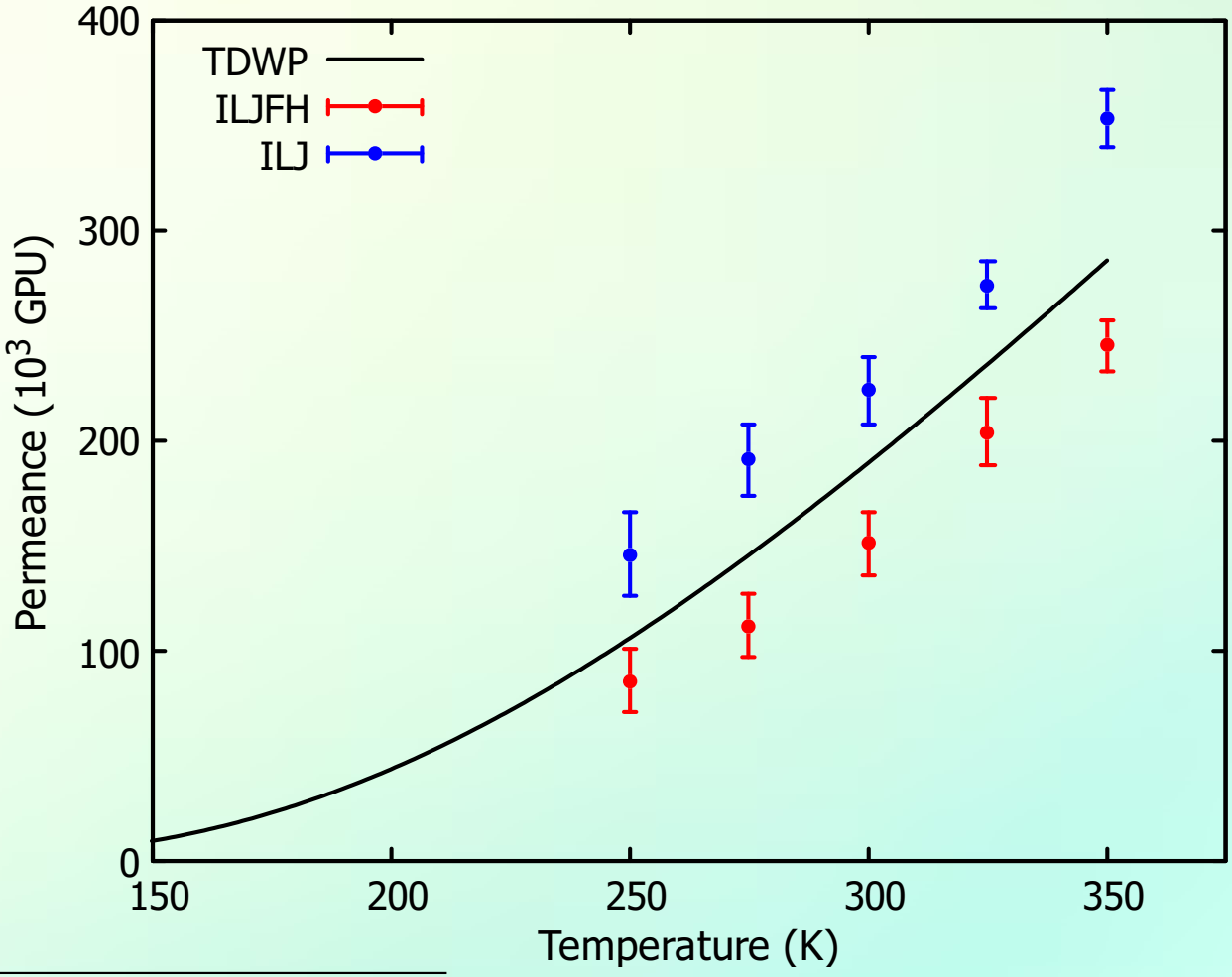
  - AIREBO (C-C)



[2] Rodríguez *et al.* *Surf. Interfaces* 95, 109659 (2026)

TDWP *vs* MD (Quantum *vs* Classical) for fixed GDY (FIX)

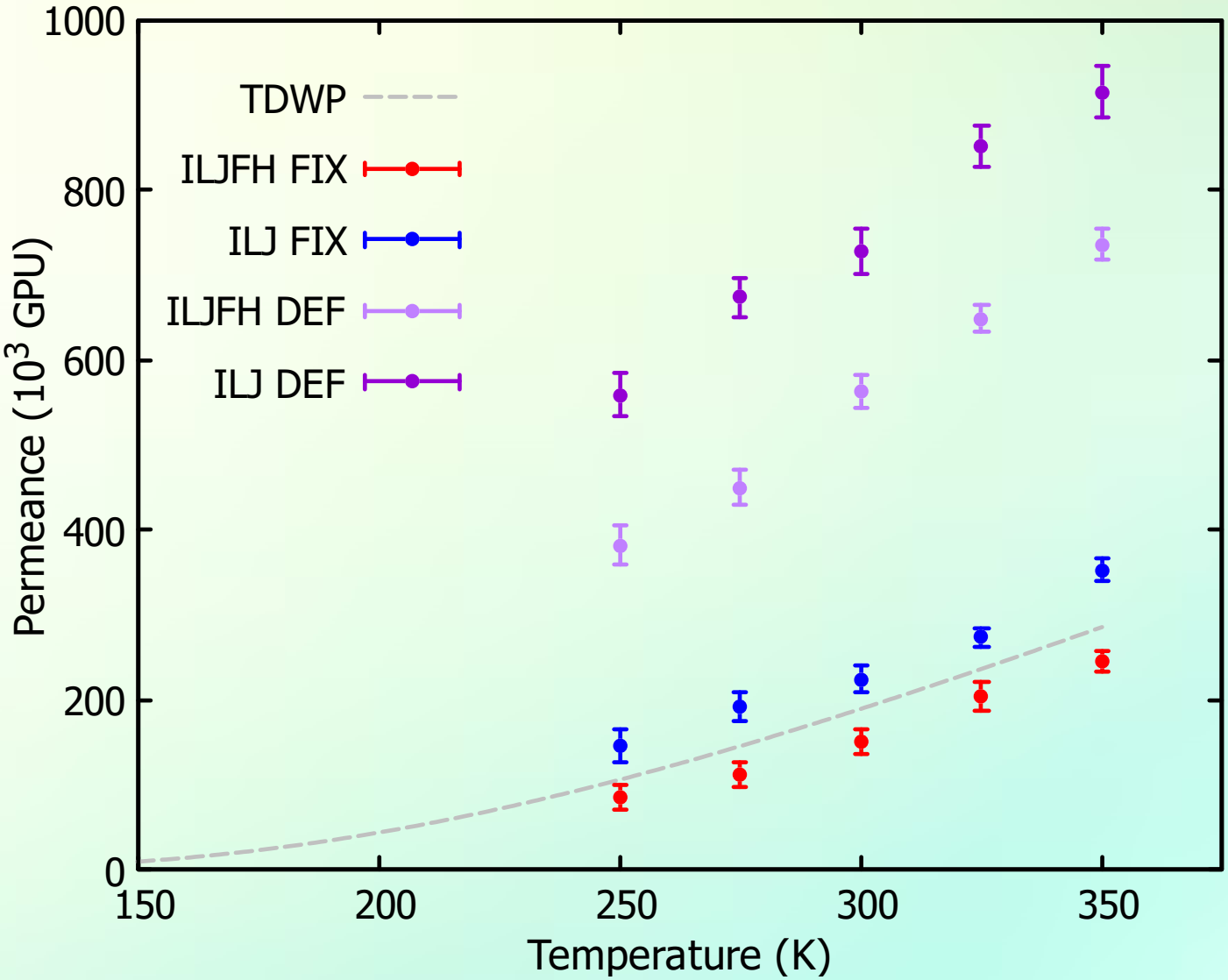
- Quantum effects, even at room temperature
- ILJFH overestimation
- ILJ and ILJFH provide upper and lower bounds



[2] Rodríguez *et al.* *Surf. Interfaces* 95, 109659 (2026)

FIX *vs* DEF

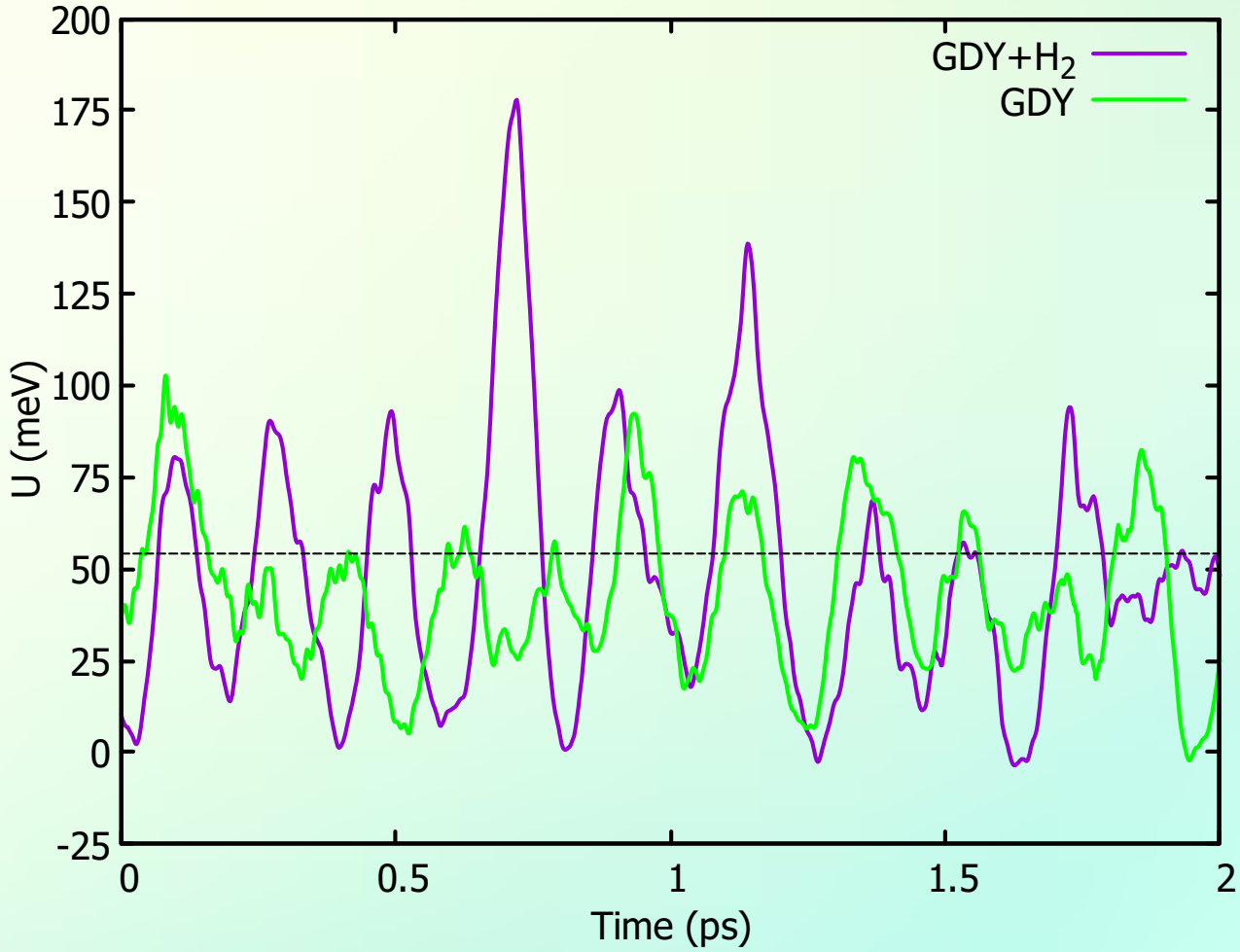
- Huge permeance increase for deformable membrane (DEF)



[2] Rodríguez *et al.* *Surf. Interfaces* 95, 109659 (2026)

FIX *vs* DEF → Potential barrier of the minimum energy path as the membrane vibrates

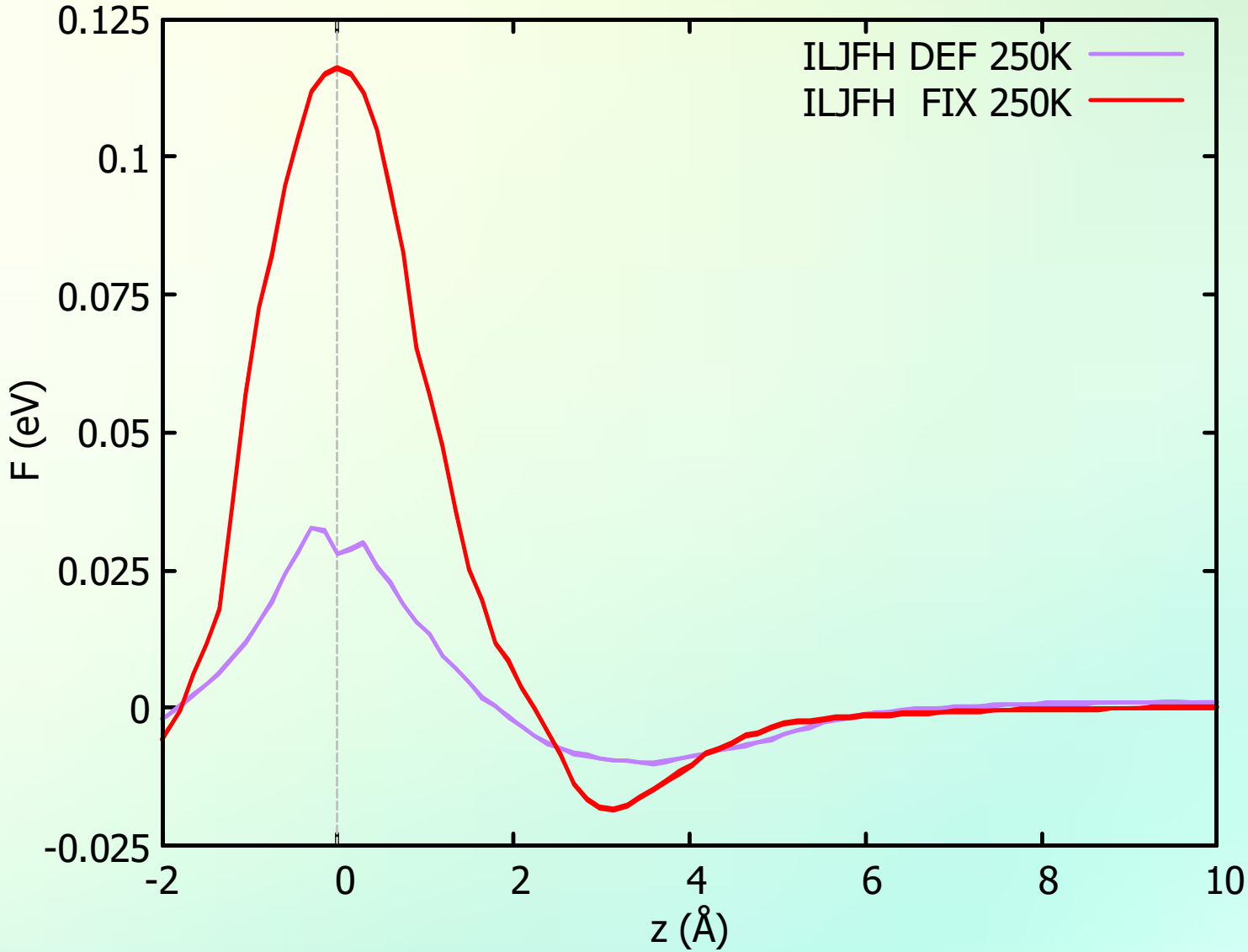
- Pore openings enhance transport



[2] Rodríguez *et al.* *Surf. Interfaces* 95, 109659 (2026)

FIX *vs* DEF → Free energy / PMF

- Great decrease in the free energy barrier for DEF



[2] Rodríguez *et al.* *Surf. Interfaces* 95, 109659 (2026)

Summary

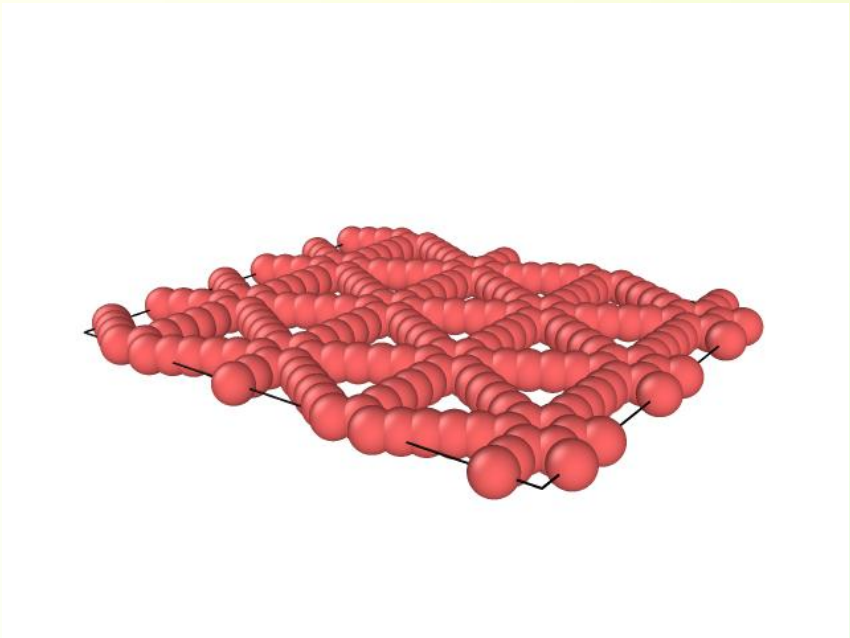
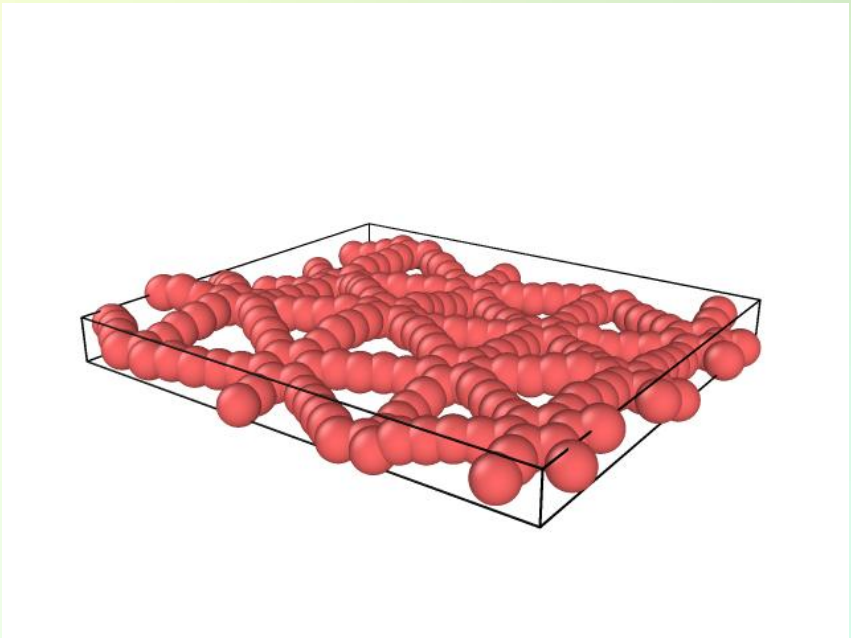
- Quantum-mechanical calculations of permeances for GDY-H_2
- Comparison of MD (classical) with TDWP (quantum)
- Significant role of membrane vibrations

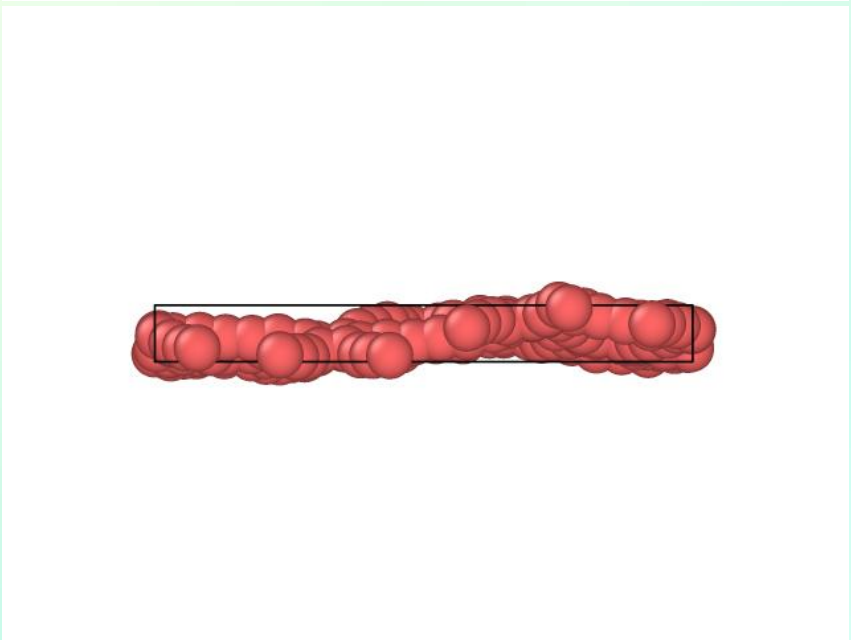
Acknowledgments

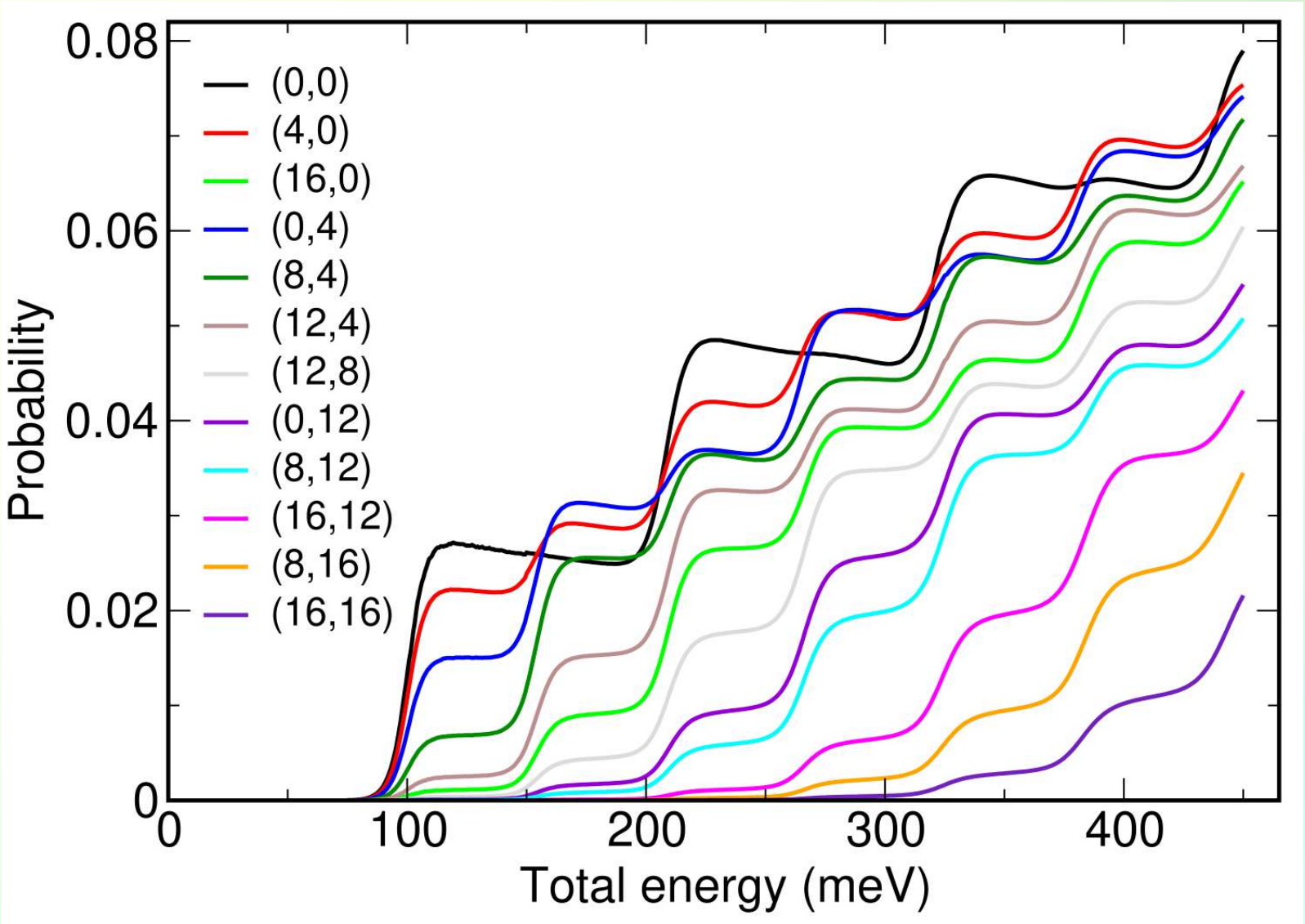
Huge thanks to my supervisors from INTERMOL (IFF-CSIC)

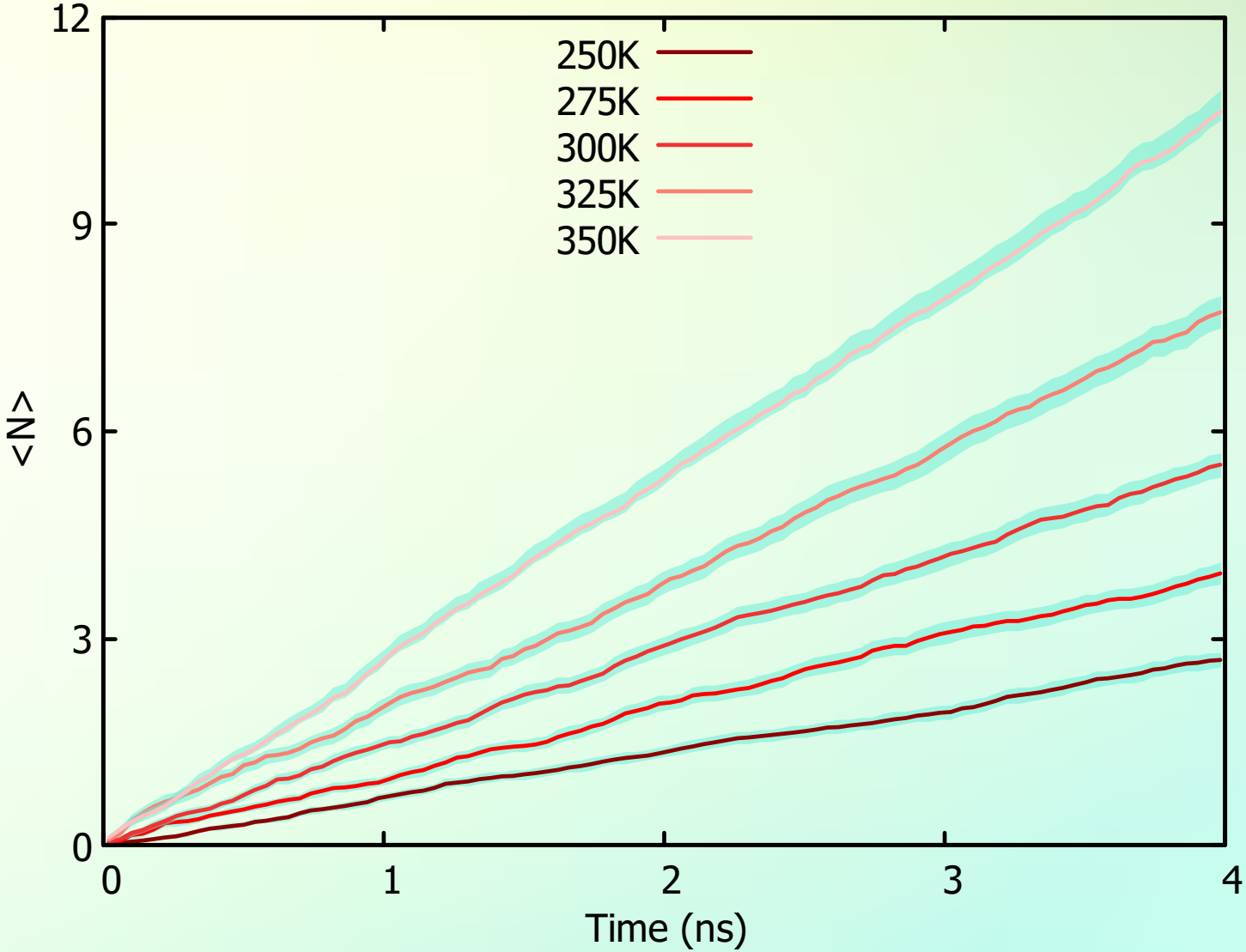
- Marta I. Hernández
- José Campos-Martínez

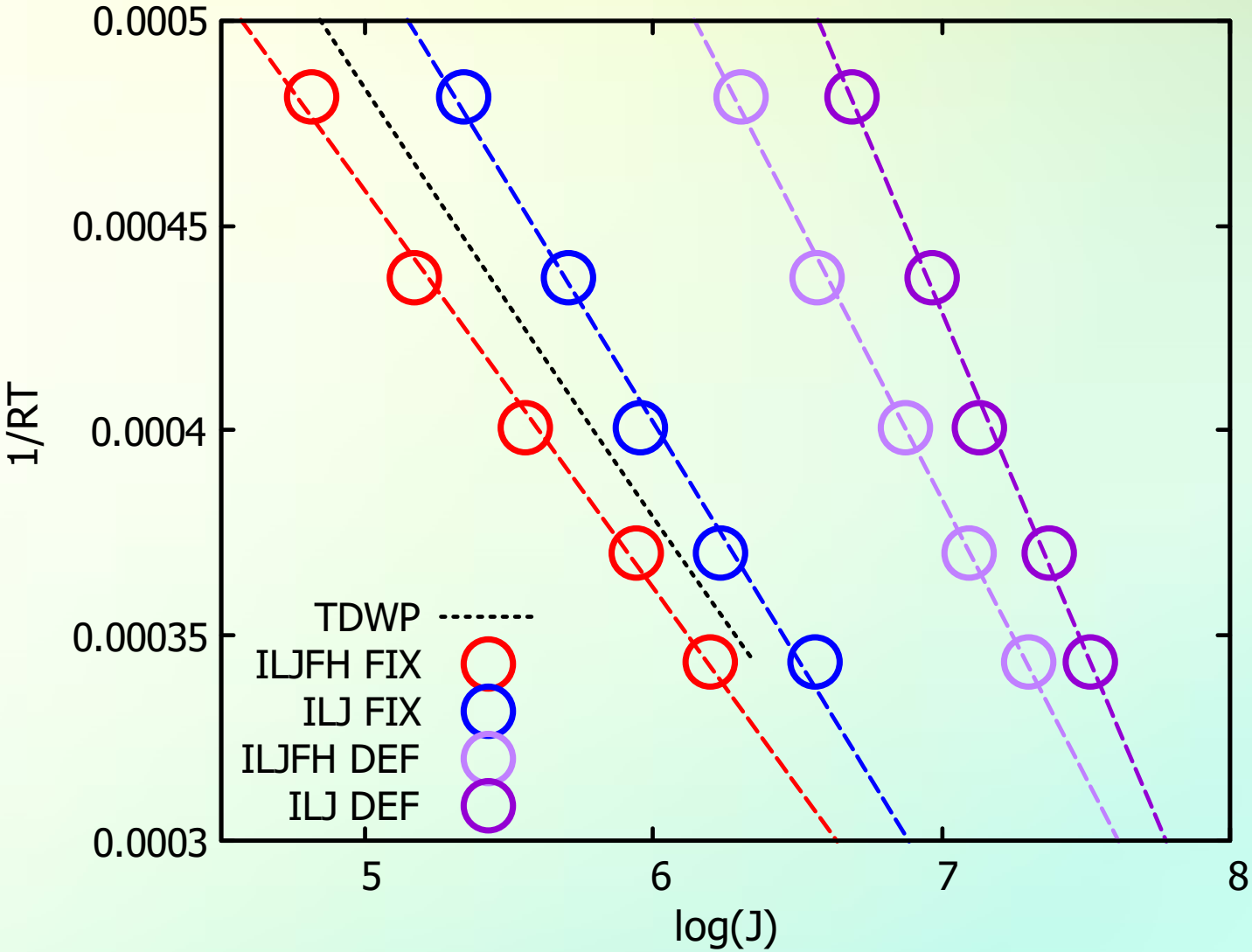

That's all Folks!

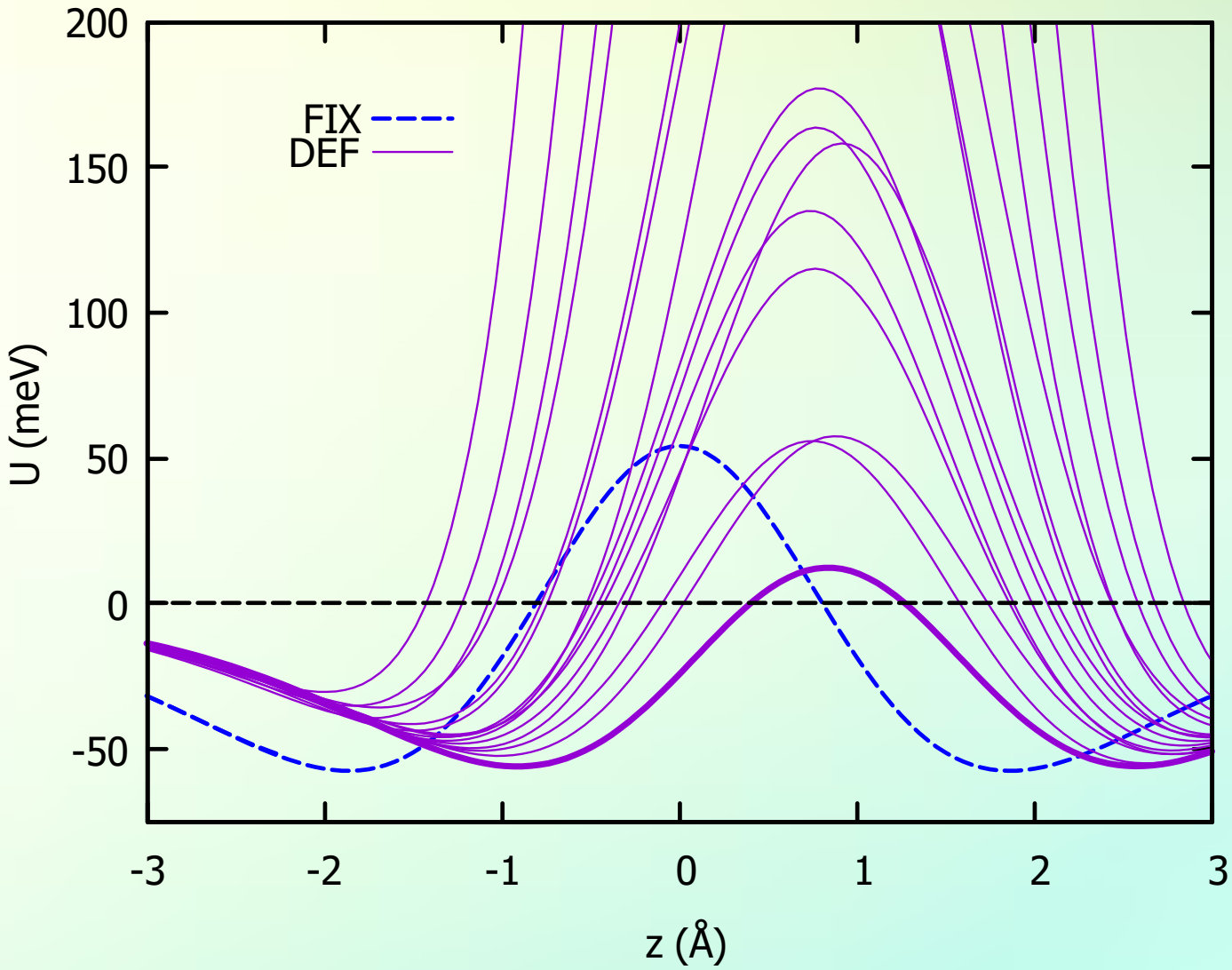



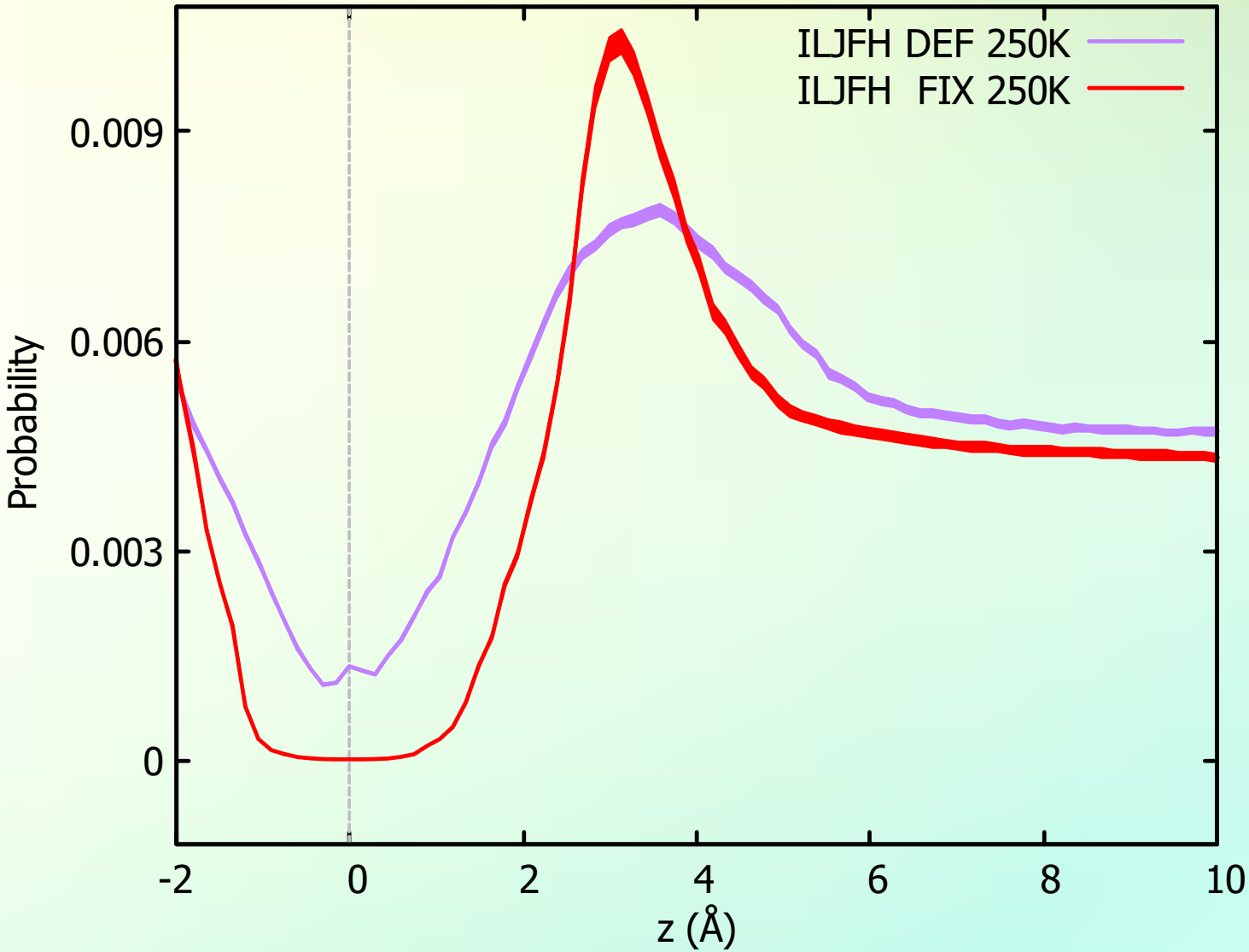



0.08
0.06
0.04
0.02
0
Probability
0
100
200
300
400
Total energy (meV)
(0,0)
(4,0)
(16,0)
(0,4)
(8,4)
(12,4)
(12,8)
(0,12)
(8,12)
(16,12)
(8,16)
(16,16)


250K
275K
300K
325K
350K
<N>
Time (ns)
0
3
6
9
12
0
1
2
3
4


0.0005
0.00045
0.0004
0.00035
0.0003
1/RT
5
6
7
8
log(J)
TDWP
ILJFH FIX
ILJ FIX
ILJFH DEF
ILJ DEF


200
150
100
50
0
-50
U (meV)
FIX
DEF
-3
-2
-1
0
1
2
3
z (Å)


ILJFH DEF 250K
ILJFH FIX 250K
Probability
0.009
0.006
0.003
0
-2
0
2
4
6
8
10
z (Å)